

# *Quiz 5 Reflection*

SCIE 001 Physics

Noah Virjee

45515863

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# 1. Hockey Physicist

## 1.1. Specify Question

Question 4: Hockey Physicist

1

2

Two identical pucks are laying flat on an icy surface and both are pulled via a thin string by a hockey player (who is a physicist at heart). The hockey player pulls with the same constant horizontal force  $F$  on both pucks for 2 seconds. In the first scenario, the thin string is wound around the outside of the puck, while in the second it is attached to its center.

a) Which puck achieves a higher speed?

Puck #2

×

0%

b) In which scenario does the hockey player do more work as they pull on the string?

Scenario #2

×

0%

This question is complete and cannot be answered again.

Correct answer

Same for both

Scenario #1

## 1.2. Diagnosis Phase

I incorrectly thought the energy imparted by the player would be the same in both cases.

After the pull, the first puck has rotational and linear kinetic energy, while the second puck only has linear kinetic energy. If the energy was the same, puck 2 would move faster linearly, while puck one would be spinning while moving slower.

I then mistakenly assumed that since work is force  $\times$  distance, and force is constant, puck two would travel farther since it was moving faster, and therefore would have had more work done on it.

Words: 92

## 1.3. Correction Phase

Energy is not the same, but force, mass, and radius are, so we can represent the systems.

Considering it from a momentum perspective:

$$F = \frac{dp}{dt}$$
$$p = \int F \, dt = Ft$$

Force is the same, so momentum is the same.

$$F_1 = F_2$$
$$F_1 t = F_2 t$$
$$p_1 = p_2$$
$$p_{1x} = p_{2x}$$
$$p_x = mv_x$$
$$mv_{1x} = mv_{2x}$$
$$v_{1x} = v_{2x}$$

So speed is the same.

For the work, we must consider the amount of rope pulled:

For the second puck:

$$F = ma$$
$$a = \frac{F}{m}$$
$$a = \frac{dv}{dt}$$
$$v_2 = \int a \, dt = \int \frac{F}{m} \, dt = \frac{Ft}{m}$$

Work is equal to force times distance:

$$v = \frac{dx}{dt}$$
$$x_2 = \int v_2 \, dt = \int \frac{Ft}{m} \, dt = \frac{Ft^2}{2m}$$
$$x_2(2) = \frac{F2^2}{2m} = \frac{2F}{m} = x_{\text{pulled}}$$
$$W_2 = |\vec{F} \times \vec{x}| = F(x_{\text{pulled}}) = \frac{2F^2}{m}$$

For the first puck, it is the same as the second, but we must account for the puck un-spooling.

Considering torque:

$$\tau = |\vec{r} \times \vec{F}| = rF$$
$$\tau = I\alpha$$
$$\alpha = \frac{\tau}{I} = \frac{rF}{I}$$

Treating the puck as a disk:

$$I = \frac{1}{2}mr^2$$
$$\alpha = \frac{rF}{\frac{1}{2}mr^2} = 2Fm^{-1}r^{-1}$$
$$\alpha = \frac{d\omega}{dt}$$
$$\omega = \int \alpha \, dt = \int 2Fm^{-1}r^{-1} \, dt = 2Fm^{-1}r^{-1}t$$
$$v_{\text{un-spooling}} = \omega r = 2Fm^{-1}r^{-1}tr = 2Fm^{-1}t$$

So the rope velocity is:

$$v_{\text{un-spooling}} = 2\left(\frac{Ft}{m}\right)$$
$$v_{\text{pulling}} = v_{\text{linear}} + v_{\text{un-spooling}}$$
$$v_{\text{pulling}} = \frac{Ft}{m} + 2\left(\frac{Ft}{m}\right) = \frac{3Ft}{m} \quad \text{with } v_{\text{linear}} = \frac{Ft}{m} \text{ from 2}$$
$$x_{\text{pulled}} = \int v_{\text{pulling}} \, dt = \int \frac{3Ft}{m} \, dt = \frac{3Ft^2}{2m}$$
$$x_{\text{pulled}}(2) = \frac{3F2^2}{2m} = \frac{6F}{m}$$
$$W_1 = |\vec{F} \times \vec{x}| = F(x_{\text{pulled}}) = \frac{6F^2}{m}$$
$$\frac{6F^2}{m} > \frac{2F^2}{m}$$
$$\therefore W_1 > W_2$$

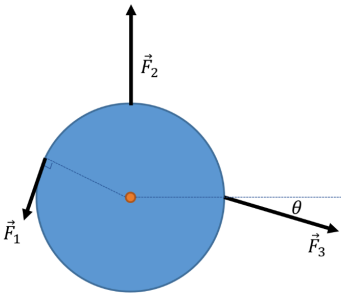
Therefore, scenario #1 has more work done.

Words: 98

## 2. Angular Velocity of Disk

### 2.1. Specify Question

Question 5: Angular Velocity of Disk



A solid uniform disk with mass  $M = 10\text{ kg}$  and radius  $R = 30\text{ cm}$  is attached to a fixed axle going through its center of mass. It is initially at rest when the following forces  $F_1 = 16\text{ N}$ ,  $F_2 = 20\text{ N}$  and  $F_3 = 32\text{ N}$  are applied to it. If  $\theta = 30^\circ$ , what is the disk's angular velocity after  $2\text{ s}$  (in  $\text{rad/s}$ )?

Assume that "out of the page" is positive and give your answer to 2 decimal places.

$\omega =$

42.666 rad / s

?

✖ 0%

This question is complete and cannot be answered again.

Correct answer

$\omega = 0.0$  radian / second

### 2.2. Diagnosis Phase

I misread the diagram and mistakenly thought that the  $\vec{F}_3$  was pointing up instead of down so when I went to calculate velocity I got:

$$\vec{\tau} = \vec{F} \times \vec{R}$$
$$\tau_1 = F_1 R \sin(90^\circ) = (16\text{N})(0.3\text{m})(1) = 4.8\text{ Nm}$$
$$\tau_2 = F_2 R \sin(180^\circ) = (20\text{N})(0.3\text{m})(0) = 0\text{ Nm}$$

I incorrectly thought that  $\vec{F}_3$  was pointing up, so I calculated its torque using  $\sin(180^\circ - 30^\circ)$  :

$$\tau_3 = F_3 R \sin(180 - 30^\circ) = (32\text{N})(0.3\text{m})(0.5) = 4.8\text{ Nm}$$
$$\tau_{\text{total}} = \tau_1 + \tau_2 + \tau_3 = 9.6\text{ Nm}$$
$$\tau = I\alpha$$

As a disk:

$$I = \frac{1}{2}mr^2$$
$$\alpha = \frac{\tau}{I} = \frac{2\tau}{mr^2}$$
$$\alpha = \frac{d\omega}{dt}$$
$$\omega = \int \alpha dt = \int \frac{2\tau}{mr^2} dt = \frac{2\tau t}{mr^2}$$
$$\omega(2) = \frac{2(9.6)(2)}{(10)(0.3)^2} = 42.67\text{ rad / s}$$

I incorrectly got:

$$\omega(2) = 42.67\text{ rad/s}$$

Words: 43

### 2.3. Correction Phase

Reading the graph correctly:

$$\vec{\tau} = \vec{F} \times \vec{R}$$
$$\tau_1 = F_1 R \sin(90^\circ) = (16\text{N})(0.3\text{m})(1) = 4.8\text{ Nm}$$
$$\tau_2 = F_2 R \sin(180^\circ) = (20\text{N})(0.3\text{m})(0) = 0\text{ Nm}$$

Interpreting the direction of  $\vec{F}_3$  properly, its torque must be calculated with  $\sin(180^\circ + 30^\circ)$

$$\tau_3 = F_3 R \sin(180 + 30^\circ) = (32\text{N})(0.3\text{m})(0.5) = -4.8\text{ Nm}$$
$$\tau_{\text{total}} = \tau_1 + \tau_2 + \tau_3 = 0\text{ Nm}$$
$$\tau = I\alpha$$

As a disk:

$$I = \frac{1}{2}mr^2$$
$$\alpha = \frac{\tau}{I} = \frac{2\tau}{mr^2}$$
$$\alpha = \frac{d\omega}{dt}$$
$$\omega = \int \alpha dt = \int \frac{2\tau}{mr^2} dt = \frac{2\tau t}{mr^2}$$
$$\omega(2) = \frac{2(0)(2)}{(10)(0.3)^2} = 0\text{ rad/s}$$

Therefore at 2 seconds the angular velocity is zero radians per second.

Words: 30